AI Study Planner

Multi-Step Prompt Workflow — Personalized Learning Pipeline

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Project Overview

An AI-powered study planning system that takes a student's learning goal, understands the required topics, retrieves useful study resources, creates a personalized schedule, checks the plan for missing or unrealistic tasks, and presents the final plan with progress and confidence information. The workflow below divides the system into prompt-driven stages that can be demonstrated and defended stage-by-stage in a viva.

Step 1 — Study Goal Understanding Prompt

Runs first on the student's raw request to identify the learning objective, available time, priority topics, and expected outcome.

```
You will receive a student study request. Do not create the schedule yet.
1. Identify the main learning goal.
2. Extract the subjects or topics that must be covered.
3. Identify available study time and deadline if provided.
4. List 2-3 priority areas for planning.

Student Request: "{student_request}"

Output as: GOAL: ... | TOPICS: ... | TIME: ... | PRIORITIES: ...
```

Step 2 — Resource Retrieval Prompt / Call

Uses the identified topics to retrieve the most relevant learning materials from a document store, notes repository, or resource index.

```
Retrieve the top {k} study resources most relevant to: {topics}.
Return each resource with its source ID, title, and relevance score.
Do not invent resources or alter the retrieved resource details.
```

Step 3 — Personalized Plan Generation Prompt

Creates a study plan using only the student's stated requirements and the retrieved resources.

```
Create a personalized study plan using ONLY the student requirements and resources below.
Balance the topics according to priority and available time.
Include study tasks, estimated duration, revision, and practice where appropriate.
If there is not enough information to build a reliable plan, say so instead of guessing.

{retrieved_resources}
Student Request: "{student_request}"

Output: DAY/TIME | TOPIC | TASK | DURATION | RESOURCE ID
```

Step 4 — Study Plan Quality Checking Prompt

Checks each planned task against the student's requirements and available resources to identify unsupported, missing, duplicated, or unrealistic items.

```
For each item in the study plan below, check whether it is supported by the student requirements
and resources.

{study_plan}
{retrieved_resources}

For each task output: VALID, NEEDS REVISION, or UNSUPPORTED with a one-line justification.
Also identify any important topic from the request that is missing from the plan.
```

Step 5 — Confidence Scoring & Adjustment Prompt

Aggregates the quality-check results into an overall plan confidence score and decides whether the plan can be accepted or needs revision.

```
Given these task-level verdicts: {verdict_list}

Compute:
- Overall confidence score (0-100%) = valid tasks / total tasks
- FLAG = "REVISION NEEDED" if any task is UNSUPPORTED or an important topic is missing,
else "READY"

Output: SCORE: {{score}}% | FLAG: {{flag}}
```

Step 6 — Final Study Plan & Progress Reporting

The final display layer presents the verified schedule, resources, confidence score, and areas that need attention.

```
Render to the student:
1. The final study schedule with daily tasks and durations
2. Topics covered and resources used
3. Confidence score and readiness flag
4. Tasks requiring revision, if any
5. Progress checklist for tracking completed tasks

Log the request, generated plan, verdicts, score, and final status for later evaluation.
```

Flow Summary

Stage	Prompt / Step	Key Output
Student request	Step 1 — Goal Understanding	Goal + topics + priorities
Resource store	Step 2 — Resource Retrieval	Relevant study resources
Plan creation	Step 3 — Plan Generation	Personalized study schedule
Quality check	Step 4 — Plan Checking	Task-level verdicts

AI Study Planner — Workflow Continuation

Scoring, Delivery and Core Principle

Stage	Prompt / Step	Key Output
Quality check	Step 4 — Plan Checking	Task-level verdicts
Scoring	Step 5 — Confidence Scoring	Score % + readiness flag
Delivery	Step 6 — Final Reporting	Plan + resources + progress

Core Principle

A study plan should be generated from the student's actual goals, time constraints, and retrieved learning resources.

The planner checks the generated tasks before presenting the schedule, so unsupported or missing study activities are identified instead of being silently included.

Student Details

Name	M.Rehana
Registration Number	99230040655
Project Title	AI Study Planner
Workflow Type	Multi-Step Prompt Workflow